D 52834 **(Pages : 2)** **Name.....**

Reg. No.....

FIRST SEMESTER M.Sc. DEGREE (REGULAR/SUPPLEMENTARY) EXAMINATION, NOVEMBER 2023

(CBCSS)

Microbiology

MBG 1C 02—BIOPHYSICS AND INSTRUMENTATION

(2019 Admission onwards)

Time : Three Hours Maximum : 30 Weightage

Part A (Short Answer Types)

Answer in two **or** *three sentences.*
Answer any **four** *out of six questions.*
Each question carries 2 *weightage.*

1. NMR Spectroscopy.
2. Gibbs free energy.
3. Alpha helix structure.
4. Isoelectric focusing.
5. Dialysis.
6. Flow cytometry.

(4 × 2 = 8 weightage)

Part B (Short Essay Types)

Answer any **four** *out of six questions.*
Each question carries 3 *weightage.*

7. Give brief account on Ramachandran Plot.
8. Give details on Beer-Lamberts Law and its applications.
9. Give details on DNA-Drug interactions.
10. Give brief account on Radioactive isotopes.

Turn over

11. Describe on various Column chromatographic techniques.

12. Differentiate SEM and TEM.

(4 × 3 = 12 weightage)

Part C (Essay Types)

Answer any **two** *out of four questions, question weightage is* 5.

13. Describe about electrophoresis and types.

14. Describe about various DNA-Protein interactions.

15. Discuss the working principle of various Spectroscopies.

16. Describe in detail about protein families.

(2 × 5 = 10 weightage)